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EDUCATION

Shandong University, Qingdao

2024 – 2028

Institute of Frontier and Interdisciplinary Science · class of 2024, expected 2028

Coursework: theoretical mechanics, electrodynamics, thermodynamics and statistical physics, quantum mechanics, advanced quantum mechanics, quantum field theory, particle physics, general relativity.

RESEARCH INTERESTS

Light-front QCD. Light-cone quantisation and DLCQ: the trivial vacuum, and where the difficulty moves instead — Fock truncation and the zero modes the $A^+=0$ gauge leaves behind; the holographic link from confinement scale to bound-state equation.

Hadron structure from Fock wave functions. The Fock-basis expansion of baryons, where a transverse-momentum distribution is a spectator-integrated amplitude normalised under $\Sigma x_i=1$, $\Sigma \mathbf{k}_{\perp i}=0$.

TMD factorisation. Collins–Soper evolution and its non-perturbative kernel; the SIDIS structure-function dictionary (Sivers, Collins, Boer–Mulders); the Drell–Yan sign reversal; SIDIS + DY global fits.

RESEARCH DIRECTIONS & WORK IN PROGRESS

- **Light-front quantisation (PH 01).** Multiplicative kinematics and a simple vacuum push every difficulty into the Fock-space truncation. I work on where it can be trusted: DLCQ matrices with a matched UV cutoff; zero modes once the gauge is fixed.
- **Baryon light-cone Fock expansion (PH 02).** Three-quark sector first, then five. The checks are normalisation ($\Sigma_n \int |\psi_n|^2=1$, $\int dx\, f_1$ equal to the valence count) and the mass constraint $P^+P^- - P_\perp{}^2 = M^2$.
- **Proton structure — DIS / SIDIS / DY (PH 03).** TMD factorisation for $P_{hT} \ll Q$ with a Y-term at large transverse momentum, Collins–Soper evolution with a non-perturbative kernel, Gaussian versus power-law tails for $\langle k_\perp{}^2 \rangle$. The 3D distribution holds only once all three agree.

OPEN QUESTIONS

- Does convergence come from restricting $k_\perp$ or from the number of Fock components?
- Do DIS, SIDIS and DY disagree on $\langle k_\perp{}^2 \rangle$ — sign reversal, or an evolution effect?
- In which Q^2 does a Gaussian width stop being defensible, i.e. where do non-Gaussian tails take over?
- Can a kernel fitted to Drell–Yan alone predict the SIDIS P_{hT} spectrum with no new parameters?

TECHNICAL SKILLS

LANGUAGES	Python, C++, Bash, Git
SYMBOLIC	Mathematica for algebra; SymPy — mathkit, an open-source skill I maintain; numerical integration and fitting
THEORY	Light-front quantisation, Fock-basis expansion, TMD factorisation and evolution, PDF and TMD fits, perturbative calculations
TOOLING	LaTeX and TikZ, reproducible notes, Linux and HPC

LANGUAGES

Chinese (native) · English (working proficiency).